A.4 SYNTHETIC DATA GENERATION DETAILS

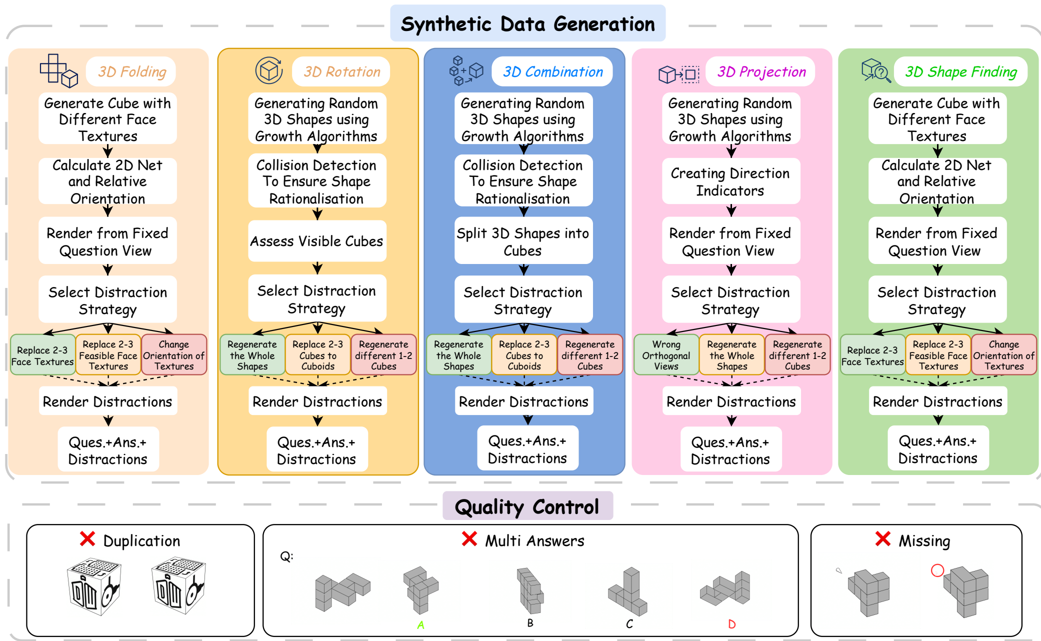

Figure 4: Synthetic Data Generation and Quality Control.

This section provides detailed procedural algorithms (pseudocode) and visual examples for the automated generation of our five core 3D spatial reasoning tasks. Each algorithm is designed for verifiability and incorporates sophisticated, task-specific strategies for generating plausible distractors.

Synthetic data generation employs Blender 4.4.0 on Apple Silicon M4. Some texture icons © Icons8 — under Universal Multimedia License. Task details, pseudocode[2], and examples for synthetic data generation are outlined below:

Algorithm 1: GENERATE3DROTATIONQUESTION

Input: question id `q_id`, list of isometric camera presets iso_views, number of distractors n
$seed \leftarrow$ Hash(`q_id`)
SetRandomSeed($seed$); ClearScene()
$orig \leftarrow$ CreateCombinationShape($cells \in [5, 15]$, *rectangularPrisms*=True, $seed$)
$qView \leftarrow$ FindBestView($orig$, iso_views)
SetCamera($qView$, *jitter*=True)
RenderImage(`q_id`$\_Q$)
$ansView \leftarrow$ ChooseDifferentView(iso_views, exclude=$qView$)
SetCamera($ansView$, *jitter*=True)
RenderImage(`q_id`$\_A0$)
for $i \leftarrow 1$ **to** n **do**
  $difficulty \leftarrow i/n$ {Higher $i \Rightarrow$ harder}
  $dShape \leftarrow$ GenerateDistractor($orig$, $difficulty$, $seed + i$)
  $dView \leftarrow$ RandomChoice(iso_views)
  SetCamera($dView$, *jitter*=True)
  RenderImage(`q_id`$\_A\{i\}$)
end for
SaveMetadata({`q_id`, $qView$, $ansView$, $seed$})

[2]All functions referenced in the code listings are project-specific utility routines; their full implementations will be provided in the accompanying public code repository.

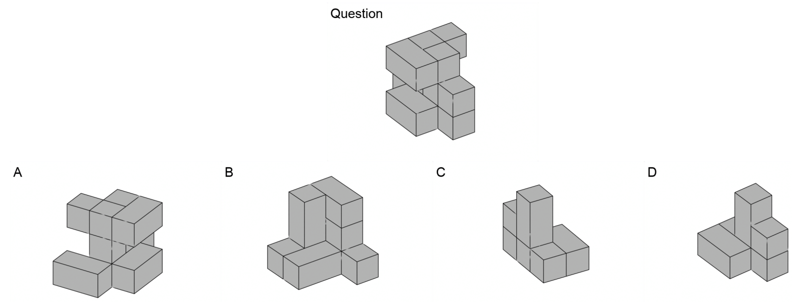

Figure 5: Synthetic 3D Rotation Data Example.

3D Rotation The 3D rotation matching task is designed to assess the ability to mentally rotate a three-dimensional object and recognize it from a different angle.

The process begins by generating a complex 3D shape composed of multiple cubes or rectangular prisms. This shape is then rendered from an optimal viewpoint to create the "question" image. This viewpoint is chosen to maximize the number of visible parts, ensuring a clear presentation of the object.

Next, a set of "answer" options is generated:

The Correct Answer: This is created by rendering the original shape from a new viewpoint, different from the one used for the question image. This requires the participant to recognize that it is the same object, despite the change in perspective.

Distractors: These are generated by creating new shapes that are slightly different from the original one. Each distractor is then rendered from a different viewpoint. These are designed to confuse the participant by presenting options that are visually similar but structurally incorrect. The final output consists of the question image, one correct answer image, and several distractor images, along with a metadata file containing all the generation parameters to ensure reproducibility.

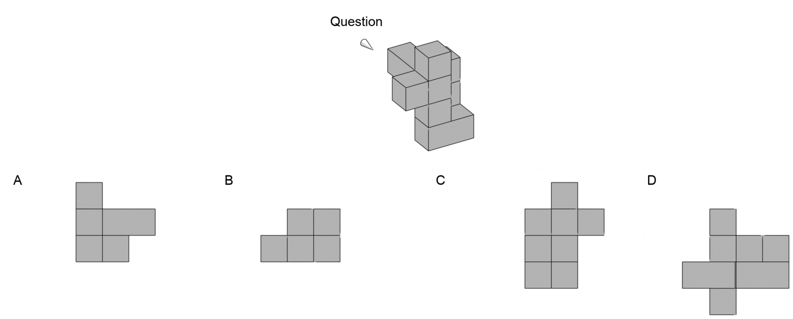

Figure 6: Synthetic 3D Projection Data Example.

3D Projection The 3D projection task evaluates a person's ability to interpret a 3D object from an isometric perspective and then identify its correct 2D orthographic projection from a set of options.

The process starts by generating a complex 3D shape. A "question" image is then created by rendering this 3D shape from an optimal isometric viewpoint. A visual cue, typically an arrow, is included in the question image to indicate the direction from which the orthographic projection should be imagined (e.g., "top-down," "front," or "side" view).

A set of options is then generated:

The Correct Answer: This is the true 2D orthographic projection of the 3D shape as seen from the direction indicated by the arrow in the question image.

Distractors: These are incorrect 2D projections. They are generated in a few ways:

Algorithm 2: GENERATE3DPROJECTIONQUESTION

```
Input: q_id, ortho_views = {top, front, right, left, bottom, back}, iso_views, number of distractors n
seed ← Hash(q_id)
SetRandomSeed(seed); ClearScene()
shape ← CreateCombinationShape(seed=seed)
qView ← FindBestView(shape, iso_views)
targetView ← RandomChoice(ortho_views)
indicator ← CreateViewIndicator(direction=targetView)
SetCamera(qView)
RenderImage(q_id_Q); Delete(indicator)
SetCameraOrtho(targetView)
RenderImage(q_id_A0)
for i ← 1 to n do
    if Random() < 0.7 then
        dShape ← GenerateDistractor(shape, difficulty=0.3 + 0.7 · i/n, seed+i)
        dView ← targetView
    else
        dShape ← shape
        dView ← ChooseDifferentView(ortho_views, exclude=targetView)
    end if
    ApplyScene(dShape)
    SetCameraOrtho(dView)
    RenderImage(q_id_A{i})
end for
SaveMetadata({q_id, seed, qView, targetView})
```

Incorrect Projections: These are valid orthographic projections but from the wrong viewpoint (e.g., a "side" view when the "top-down" view was asked for).

Slightly Altered Shapes: These are 2D projections of shapes that are subtly different from the original 3D shape, testing attention to detail. The participant must select the 2D image that accurately represents the specified orthographic projection of the 3D object shown in the question.

Algorithm 3: GENERATE3DCOMBINATIONQUESTION

```
Input: q_id, iso_views, number of distractors n
seed ← Hash(q_id)
SetRandomSeed(seed); ClearScene()
master ← CreateCombinationShape(seed=seed, complexity=medium)
qView ← FindBestView(master, iso_views); SetCamera(qView)
RenderImage(q_id_Q)
oppView ← OppositeView(qView); SetCamera(oppView)
RenderImage(q_id_Q_opp)
components ← DeconstructShape(master)
ArrangeComponentsGrid(components, gap=2)
SetCamera(GlobalOverview)
RenderImage(q_id_A0)
for i ← 1 to n do
    comp ← RandomChoice(components)
    dComp ← CreateDistractorComponent(comp, variation=i/n, seed+i)
    ReplaceComponent(comp, dComp)
    RenderImage(q_id_A{i})
    RestoreComponent(comp)
end for
SaveMetadata({q_id, seed, mainView:qView, oppView:oppView})
```